Derivatives

1. Motivation

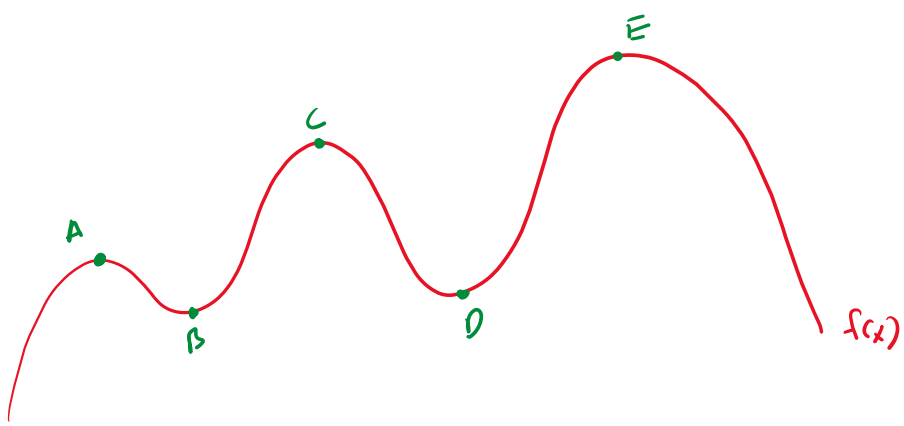
If $f(x)$ is a quadratic then the graph is

and we can find the maximum or minimum of $f(x)$.

what if $f(x)$ is not a quadratic ?

How can we find the max / min of $f(x)$?

we use " Derivatives" to do that.

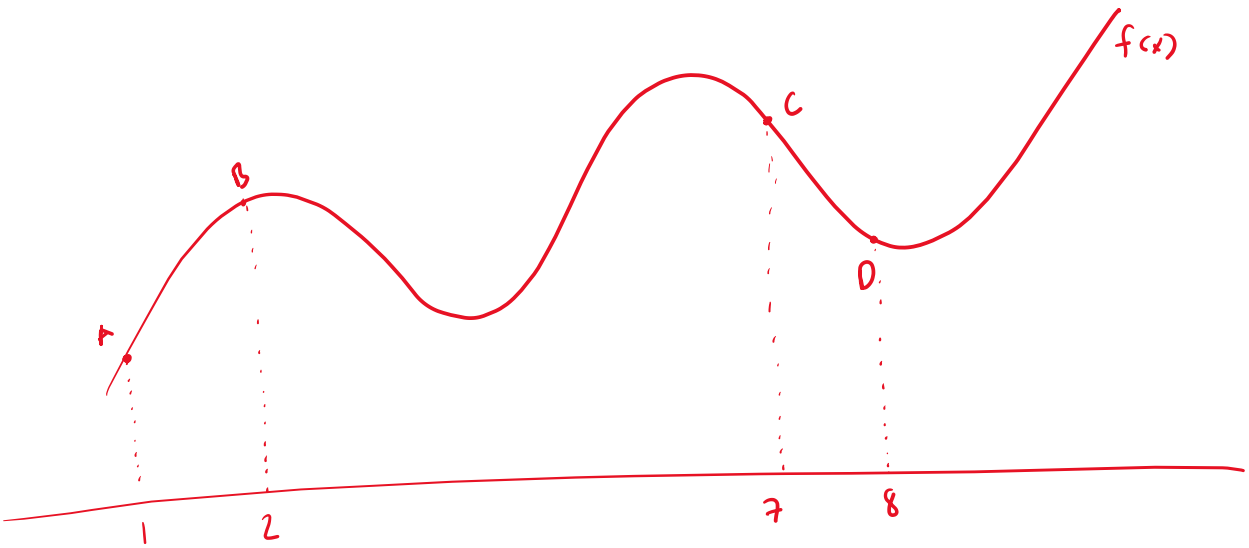


using " Derivatives" we can find the locations of A, B, C, D and E . Hence, we can find the max of $f(x)$.

2. The idea of “derivative”

The derivative of a function f(x) is denoted by f’(x) [reads: f prime of x]. And f’(a) measures the instant change of f(x) at the point a.

- If f’(a) >0 then f(x) is increasing at x = a.
- If f’(a) <0 then f(x) is decreasing at x = a.
- The absolute value of f’(a) is the speed of increasing or decreasing at x = a.



$f'(1) > 0$ (f(x) going up at x = 1)

$f'(2) > 0$

$f'(7) < 0$, $f'(8) < 0$

$f'(1) > f'(2)$ (f(x) going up faster at x = 1)

$|f'(7)| > |f'(8)|$ (f(x) going down faster at x = 7)

3. Calculating derivatives

We will use several "rules" to calculate the derivative of a function.

<u>Rule 1</u> : Constant Rule

If $f(x) = c$ (constant)

$\Rightarrow f'(x) = 0$

<u>OR</u> : $(c)' = 0$

<u>Rule 2</u> : Linear Rule

If $f(x) = c \cdot x$ (linear function)

$\Rightarrow f'(x) = c$

<u>OR</u> : $(cx)' = c$

Example : $(4x)' = 4$

$(2026x)' = 2026$

<u>Rule 3</u> : Power Rule

If $f(x) = x^n$ (power function)

$\Rightarrow f'(x) = n \cdot x^{n-1}$

<u>OR</u> : $(x^n)' = n \cdot x^{n-1}$

Example : Find $f'(x)$

① $f(x) = x^{10} \Rightarrow f'(x) = 10 \cdot x^{10-1}$

$\Rightarrow f'(x) = 10 \cdot x^{9}$

② $f(x) = x^{3} \Rightarrow f'(x) = 3x^{3-1}$

$f'(x) = 3x^{2}$

③ $f(x) = x^{2} \Rightarrow f'(x) = 2x^{2-1} = 2x$

④ $f(x) = \frac{1}{x^{7}} = x^{-7}$

$$\boxed{\text{Formula : } \frac{1}{x^{k}} = x^{-k}}$$

$\Rightarrow f'(x) = -7 \cdot x^{-7-1}$

$\Rightarrow f'(x) = -7x^{-8} = \frac{-7}{x^{8}}$

⑤ $f(x) = \frac{1}{x} = x^{-1}$

$\Rightarrow f'(x) = -1 \cdot x^{-1-1} = -x^{-2}$

$= \frac{-1}{x^{2}}$

Practice: Find $f'(x)$

1. $f(x) = 2026$
2. $f(x) = -7x$
3. $f(x) = \frac{1}{3}x$
4. $f(x) = x^{100}$
5. $f(x) = x^{20}$
6. $f(x) = \frac{1}{x^9}$
7. $f(x) = \frac{1}{x^{5/3}}$